

TEAM ID	SWTID-2026-4030
PROJECT TITLE	Plugging into the Future: An Exploration of Electricity Consumption Patterns
MAX MARKS	3 MARKS

6.2 – Data Connectivity

For the project "Plugging into the Future: An Exploration of Electricity Consumption Patterns", MySQL was used as the backend database to store and manage electricity consumption data efficiently.

The connectivity was established using the Flask framework and MySQL Connector/MySQL library. The dataset containing electricity consumption records was imported into MySQL tables, where data such as consumption values, dates, regions, and customer details were stored. Flask applications connected to the MySQL database using secure database credentials and executed SQL queries to retrieve, update, and analyse data.

The retrieved data was processed and displayed through the web application dashboard. This integration enabled efficient data management, faster querying, real-time data access, and seamless visualization through the embedded dashboard. MySQL ensured data integrity, scalability, and reliable storage, making it easier to perform consumption pattern analysis and generate meaningful insights for users.

Technology Stack Used:

- MySQL Database
- Flask Framework
- Python
- MySQL Connector / MySQL
- HTML, CSS, Bootstrap
- Tableau Dashboard Integration

Plugging into the Future: An Exploration of Electricity Consumption Patterns

This database connectivity forms the backbone of the project by enabling smooth storage, retrieval, and analysis of electricity consumption data. By linking smart devices, communication networks, and analytical systems, utilities and consumers can gain valuable insights into energy usage patterns, leading to smarter energy management, lower costs, and a more sustainable future.

